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LECTURE 8: Scattering amplitudes and cross-sections. The Born approximation for spherically symmetric potentials.

Scattering is one of the main ways we learn about physical systems. When a particle or wave is sent toward a target, the potential reshapes the outgoing wave, and this distortion carries information about the interaction. The key quantity that captures this information is the scattering amplitude $f(\theta, \phi)$ which multiplies the outgoing spherical wave after the interaction and contains the angles θ and ϕ defined as usual assuming that the incoming wave is along the z -direction. It tells us how much of the incoming flux is redirected into a given solid angle, and it encodes both the strength and the angular dependence of the interaction. Most measurable quantities in scattering experiments—especially the differential cross section—follow directly from it through

$$\frac{d\sigma}{d\Omega} = |f(\theta, \phi)|^2.$$

This makes $f(\theta, \phi)$ the central bridge between theory and experiment. The goal of the lecture is to understand how the scattering amplitude emerges from the asymptotic solution of the Schrödinger equation and how its structure reveals the underlying physics of the potential.

In this lecture we will first justify/prove the Lippmann-Schwinger equation. Then we proceed by

Procedure

- (i) deriving the asymptotic form in the

$$\psi^{(+)}(\mathbf{r}) \simeq \frac{1}{L^{3/2}} \left[e^{i\mathbf{k}_i \cdot \mathbf{r}} + \frac{e^{ikr}}{r} f(\theta, \phi) \right].$$

as a sum of the incoming plane wave and a spherical wave with the scattering amplitude

$$f(\theta, \phi) = -\frac{2\pi^2 m}{\hbar^2} \langle \mathbf{k}_f | \hat{T}(E_i) | \mathbf{k}_i \rangle.$$

- (ii) introducing the first Born approximation where $\hat{T}(E_i)$ is simply replaced by $\hat{V}$ for $f(\theta, \phi)$, to obtain

$$f^{(1)}(\theta, \phi) \propto -\frac{m}{\hbar^2} \int d^3r e^{-i(\mathbf{k}_f - \mathbf{k}_i) \cdot \mathbf{r}} V(\mathbf{r}) = -\frac{m}{\hbar^2} \tilde{V}(\mathbf{q}),$$

with $\mathbf{q} = \mathbf{k}_f - \mathbf{k}_i$ the momentum transfer and $\tilde{V}(\mathbf{q})$ the Fourier transform of $V(\mathbf{r})$.

- (iii) providing a few examples of computing the cross sections for spherically symmetric potentials where for elastic scattering the scattering is elastic $q = |\mathbf{k} - \mathbf{k}'| = 2k \sin \frac{\theta}{2}$, and

$$f^{(1)}(\theta) = -\frac{2m}{\hbar^2} \frac{1}{q} \int_0^\infty r V(r) \sin(qr) dr.$$

As you see, the problem can be reduced to finding the T matrix, from which one can derive the differential and total cross section. In the particular case of weak perturbation, it is then enough to compute the Fourier transform of the potential and derive everything from it. Even more straightforward is the case of weak spherically symmetric potentials, where for low enough incoming energies the first Born approximation gives an easy procedure for computing the angle dependence of the scattered states.

Derivation of the Lippmann–Schwinger equation

We begin by recalling that a scattering experiment starts in the remote past at $t_0 = -\infty$, where the particle is far away from the potential, moves freely, and therefore must be in a free eigenstate of $\hat{H}_0$.

We begin with the time-dependent Schrödinger equation

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = (\hat{H}_0 + \hat{V}) |\psi(t)\rangle,$$

together with the incoming boundary condition expressing that long before the scattering occurs the particle is free in an eigenstate $|\mathbf{k}_i\rangle$ with energy E_i .

To reorganize the problem we transform into the interaction picture via

$$|\psi_I(t)\rangle = e^{\frac{i}{\hbar} \hat{H}_0 t} |\psi(t)\rangle, \quad \hat{V}_I(t) = e^{\frac{i}{\hbar} \hat{H}_0 t} \hat{V} e^{-\frac{i}{\hbar} \hat{H}_0 t}.$$

In this picture the Schrödinger equation is simplified to

$$i\hbar \frac{d}{dt} |\psi_I(t)\rangle = \hat{V}_I(t) |\psi_I(t)\rangle.$$

Now we integrate the interaction-picture Schrödinger equation formally. This yields a Dyson-type integral equation, in which the unknown state $|\psi_I(t')\rangle$ appears under the integral sign. To ensure that the state is $|\mathbf{k}_i\rangle$ at $t' \rightarrow -\infty$, we insert an adiabatic factor $e^{\epsilon t'}$ with $\epsilon > 0$ in the formal solution

$$|\psi_I(t)\rangle = |\mathbf{k}_i\rangle - \frac{i}{\hbar} \int_{-\infty}^t dt' e^{\epsilon t'} \hat{V}_I(t') |\psi_I(t')\rangle.$$

This is not a solution; it is an integral equation because $|\psi(t')\rangle$ appears inside the integral. To obtain a stationary scattering state, we evaluate this equation at a single time at $t = 0$. At $t = 0$ the interaction and Schrödinger pictures coincide, so we define the stationary scattering state as

$$|\psi^{(+)}\rangle \equiv |\psi(0)\rangle = |\psi_I(0)\rangle.$$

Physically this means: the scattering process takes place from $t = -\infty$ to $t = 0$. After $t = 0$ the particle moves freely again. Therefore, $|\psi^{(+)}\rangle$ is the complete scattering state containing both incoming and outgoing parts.

Evaluating the integral equation at $t = 0$ gives

$$|\psi^{(+)}\rangle = |\mathbf{k}_i\rangle - \frac{i}{\hbar} \int_{-\infty}^0 dt' e^{\epsilon t'} \hat{V}_I(t') |\psi_I(t')\rangle.$$

Next, we express $|\psi_I(t')\rangle$ entirely in terms of the stationary state $|\psi^{(+)}\rangle$. This is the key step: instead of integrating over ψ , we rewrite $|\psi_I(t')\rangle$ in a fixed vector form.

The scattering state is stationary with energy E_i , hence

$$|\psi(t)\rangle = e^{-\frac{i}{\hbar} E_i t} |\psi^{(+)}\rangle.$$

Transforming to the interaction picture gives

$$|\psi_I(t')\rangle = e^{-\frac{i}{\hbar} (E_i \mathbb{1} - \hat{H}_0) t'} |\psi^{(+)}\rangle,$$

and $\hat{V}_I(t') = e^{\frac{i}{\hbar}\hat{H}_0 t'} \hat{V} e^{-\frac{i}{\hbar}\hat{H}_0 t'}$.

Insert both expressions into the integral equation to get

$$\hat{V}_I(t') |\psi_I(t')\rangle = e^{-\frac{i}{\hbar}E_i t'} e^{\frac{i}{\hbar}\hat{H}_0 t'} \hat{V} |\psi^{(+)}\rangle.$$

Substituting,

$$|\psi^{(+)}\rangle = |\mathbf{k}_i\rangle - \frac{i}{\hbar} \int_{-\infty}^0 dt' e^{\epsilon t'} e^{-\frac{i}{\hbar}E_i t'} e^{\frac{i}{\hbar}\hat{H}_0 t'} \hat{V} |\psi^{(+)}\rangle,$$

or

$$|\psi^{(+)}\rangle = |\mathbf{k}_i\rangle - \hat{K}(E_i) \hat{V} |\psi^{(+)}\rangle,$$

where

$$\hat{K}(E_i) = \frac{i}{\hbar} \int_{-\infty}^0 dt' e^{\epsilon t'} e^{-\frac{i}{\hbar}E_i t'} e^{\frac{i}{\hbar}\hat{H}_0 t'}.$$

To evaluate $\hat{K}(E_i)$, we change variables $\tau = -t'$ such that $dt' = -d\tau$, and compute this operator integral in the momentum basis.

$$\langle n | \hat{K}(E_i) | m \rangle = \delta_{nm} \frac{i}{\hbar} \int_0^{\infty} d\tau e^{-\epsilon \tau} e^{\frac{i}{\hbar}(E_i - E_n)\tau} = \delta_{nm} \frac{1}{E_i - E_n + i\epsilon}.$$

This shows that the operator is diagonal and can be casted in operator form as

$$\hat{K}(E_i) = \frac{1}{E_i \mathbb{1} - \hat{H}_0 + i\epsilon \mathbb{1}} \equiv \hat{G}(E_i).$$

This finally gives the form of the Lippmann-Schwinger that we have used:

$$|\psi^{(+)}\rangle = |\mathbf{k}_i\rangle + \hat{G}(E_i) \hat{V} |\psi^{(+)}\rangle.$$

Introducing the T-matrix

It is convenient to isolate the part resulting from scattering. To this end, we introduce the **T-matrix** via the following relation

$$\hat{T}(E_i) |\mathbf{k}_i\rangle \equiv \hat{V} |\psi^{(+)}\rangle.$$

With this definition, the Lippmann-Schwinger equation becomes

$$|\psi^{(+)}\rangle = |\mathbf{k}_i\rangle + \hat{G}(E_i) \hat{T}(E_i) |\mathbf{k}_i\rangle.$$

To derive an equation for the T-matrix, we insert the LS equation into the definition:

$$\hat{T}(E_i) |\mathbf{k}_i\rangle = \hat{V} |\psi^{(+)}\rangle = \hat{V} |\mathbf{k}_i\rangle + \hat{V} \hat{G}(E_i) \hat{V} |\psi^{(+)}\rangle.$$

Using again the definition of $\hat{T}(E_i)$, we obtain

$$\hat{T}(E_i) |\mathbf{k}_i\rangle = \hat{V} |\mathbf{k}_i\rangle + \hat{V} \hat{G}(E_i) \hat{T}(E_i) |\mathbf{k}_i\rangle.$$

Since this must hold for any incoming $|\mathbf{k}_i\rangle$, we arrive at the following equation for the T-matrix

$$\hat{T}(E_i) = \hat{V} + \hat{V} \hat{G}(E_i) \hat{T}(E_i).$$

The scattering problem then simply becomes the task to solve for $\hat{T}(E_i)$.

The scattering amplitude

Let us remind ourselves again that we work in a cubic quantization box of volume L^3 with periodic boundary conditions. The momentum eigenstates are

$$\langle \mathbf{r} | \mathbf{k} \rangle = \frac{1}{L^{3/2}} e^{i\mathbf{k} \cdot \mathbf{r}}, \quad \mathbf{k} = \frac{2\pi}{L} (n_x, n_y, n_z),$$

and satisfy

$$\sum_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}| = \mathbb{1}, \quad \langle \mathbf{k} | \mathbf{k}' \rangle = \delta_{\mathbf{k}, \mathbf{k}' }.$$

The Lippmann–Schwinger equation in the box reads

$$|\psi^{(+)}\rangle = |\mathbf{k}_i\rangle + \hat{G}(E_i) \hat{V} |\psi^{(+)}\rangle, \quad \hat{G}(E_i) = \frac{1}{E_i \mathbb{1} - \hat{H}_0 + i\epsilon \mathbb{1}}.$$

In the position representation,

$$\psi^{(+)}(\mathbf{r}) = \langle \mathbf{r} | \psi^{(+)} \rangle = \frac{1}{L^{3/2}} e^{i\mathbf{k}_i \cdot \mathbf{r}} + \int_V d^3 r' G(\mathbf{r}, \mathbf{r}'; E_i) V(\mathbf{r}') \psi^{(+)}(\mathbf{r}'),$$

where the box Green function at the elastic energy $E_i = \hbar^2 k_i^2 / (2m)$ is given by

$$G(\mathbf{r}, \mathbf{r}'; E_i) = \sum_{\mathbf{k}, \mathbf{k}'} \langle \mathbf{r} | \mathbf{k} \rangle \langle \mathbf{k} | \hat{G}(E) | \mathbf{k}' \rangle \langle \mathbf{k}' | \mathbf{r}' \rangle = -\frac{2m}{\hbar^2} \frac{1}{L^3} \sum_{\mathbf{k}} \frac{e^{i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')}}{k^2 - k_i^2 - i\epsilon}.$$

Substituting this into the Lippmann–Schwinger equation gives

$$\psi^{(+)}(\mathbf{r}) = \frac{1}{L^{3/2}} e^{i\mathbf{k}_i \cdot \mathbf{r}} - \frac{2m}{\hbar^2} \frac{1}{L^3} \sum_{\mathbf{k}} \frac{e^{i\mathbf{k} \cdot \mathbf{r}}}{k^2 - k_i^2 - i\epsilon} \int_V d^3 r' e^{-i\mathbf{k} \cdot \mathbf{r}'} V(\mathbf{r}') \psi^{(+)}(\mathbf{r}').$$

In the next step, lots of approximations are performed. We will skip these and refer the interested reader to the textbook derivation in sakurai and only point out the results and how they should be understood.

The pole of the summand at $k^2 = k_i^2$ generates, for large r (much larger with respect to the range of the potential), an outgoing spherical wave. The total wave function can then be written as an asymptotic form

$$\psi^{(+)}(\mathbf{r}) \simeq \frac{1}{L^{3/2}} \left[e^{i\mathbf{k}_i \cdot \mathbf{r}} + \frac{e^{ik_i r}}{r} f(\theta, \phi) \right].$$

This is the standard result: a unit-normalized incoming plane wave plus an outgoing spherical wave whose angular dependence is encoded in the scattering amplitude

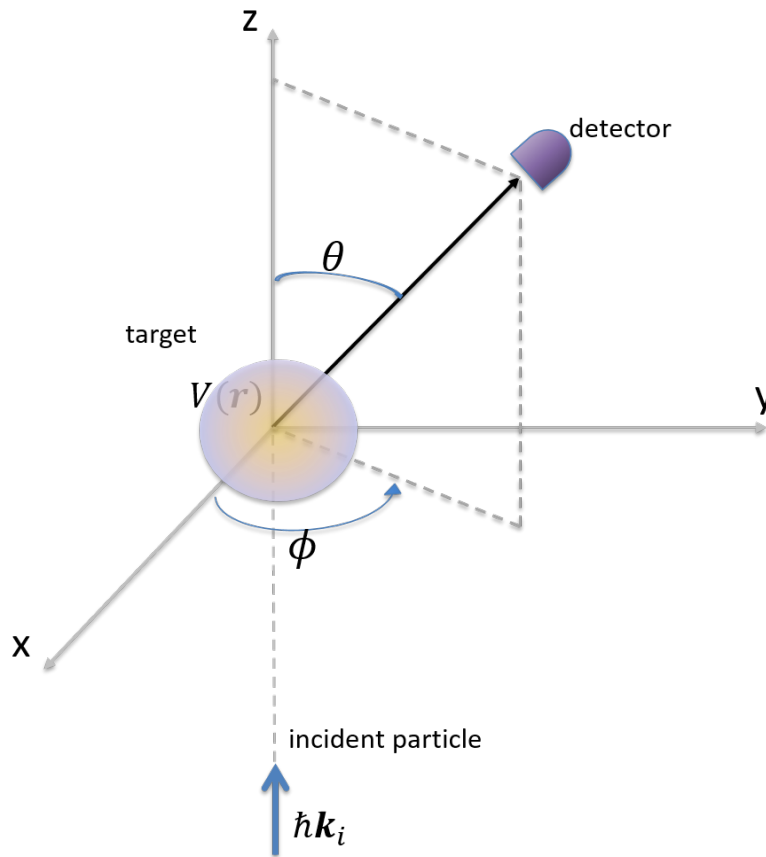
$$f(\theta, \phi) = -\frac{2m}{\hbar^2} \frac{L^{3/2}}{4\pi} \int_V d^3 r' e^{-i\mathbf{k}_f \cdot \mathbf{r}'} V(\mathbf{r}') \psi^{(+)}(\mathbf{r}'), \quad \mathbf{k}_f = k_i \hat{r},$$

the asymptotic form of the wavefunction becomes

$$\psi^{(+)}(\mathbf{r}) \simeq \frac{1}{L^{3/2}} \left[e^{i\mathbf{k}_i \cdot \mathbf{r}} + \frac{e^{ik_i r}}{r} f(\theta, \phi) \right].$$

It can be shown that the scattering amplitude can be also expressed simply as the matrix element of the T -matrix:

$$f(\theta, \phi) = -\frac{mL^3}{2\pi\hbar^2} \langle \mathbf{k}_f | \hat{T}(E_i) | \mathbf{k}_i \rangle.$$



Differential cross section from the scattered wave

To extract physical scattering information from the asymptotic wavefunction we use the probability current density. This is derived from a continuity equation involving the probability density $\rho = |\psi|^2$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0.$$

For a particle of charge q the charge current will then be $\mathbf{J} = q\mathbf{j}$. The current density can be computed from the Schrödinger equation using the continuity above to give:

$$\mathbf{j}(\mathbf{r}) = \frac{\hbar}{m} \Im[\psi^*(\mathbf{r}) \nabla \psi(\mathbf{r})],$$

read as the flow of probability per unit area per unit time.

For the incoming normalized plane wave one finds

$$j_{\text{inc}} = \frac{\hbar k}{m} \frac{1}{L^3},$$

a constant flux along the direction of $\mathbf{k}_i$.

The scattered part of the wavefunction is

$$\psi_{\text{sc}}(\mathbf{r}) = \frac{1}{L^{3/2}} \frac{e^{ikr}}{r} f(\theta, \phi).$$

At large distance this becomes a purely radial outgoing wave. Inserting it into the definition of the probability current and keeping the dominant term (the derivation will be left for the exercise session), the radial component of the scattered

current is

$$j_r(\theta, \phi) = \frac{\hbar k}{m} \frac{1}{L^3} \frac{|f(\theta, \phi)|^2}{r^2}.$$

Multiplying this radial flux by the area element $r^2 d\Omega$ gives the rate at which probability flows into the solid angle $d\Omega$:

$$\frac{dN_{sc}}{dt} = j_r r^2 d\Omega = \frac{\hbar k}{m} \frac{1}{L^3} |f(\theta, \phi)|^2 d\Omega.$$

The differential cross section is defined as the ratio of this scattered rate to the incident flux,

$$\frac{d\sigma}{d\Omega} = \frac{\frac{dN_{sc}}{dt}}{j_{inc}} = |f(\theta, \phi)|^2.$$

Thus the angular pattern of the scattered spherical wave directly encodes the measurable differential cross section.

The first Born approximation

The exact T -matrix is often difficult to compute. A natural approximation scheme is the first order in the interaction (first Born approximation), in which we replace $\hat{T}(E_i) \rightarrow \hat{V}$. Inserting this into the expression for the scattering amplitude gives

$$f^{(1)}(\theta, \phi) = -\frac{2m}{\hbar^2} \frac{L^{3/2}}{4\pi} \langle \mathbf{k}_f | \hat{V} | \mathbf{k}_i \rangle,$$

where the box-normalized plane waves satisfy

$$\langle \mathbf{r} | \mathbf{k} \rangle = \frac{1}{L^{3/2}} e^{i\mathbf{k} \cdot \mathbf{r}}.$$

To evaluate $\langle \mathbf{k}_f | \hat{V} | \mathbf{k}_i \rangle$ we insert completeness relations in the position basis:

$$\langle \mathbf{k}_f | \hat{V} | \mathbf{k}_i \rangle = \int_V d^3r d^3r' \langle \mathbf{k}_f | \mathbf{r} \rangle \langle \mathbf{r} | \hat{V} | \mathbf{r}' \rangle \langle \mathbf{r}' | \mathbf{k}_i \rangle \quad (1)$$

$$= \int_V d^3r \left(\frac{1}{L^{3/2}} e^{-i\mathbf{k}_f \cdot \mathbf{r}} \right) V(\mathbf{r}) \left(\frac{1}{L^{3/2}} e^{i\mathbf{k}_i \cdot \mathbf{r}} \right) \quad (2)$$

$$= \frac{1}{L^3} \int_V d^3r e^{-i(\mathbf{k}_f - \mathbf{k}_i) \cdot \mathbf{r}} V(\mathbf{r}). \quad (3)$$

With the momentum transfer $\mathbf{q} = \mathbf{k}_f - \mathbf{k}_i$, this becomes

$$\langle \mathbf{k}_f | \hat{V} | \mathbf{k}_i \rangle = \frac{1}{L^3} \int_V d^3r e^{-i\mathbf{q} \cdot \mathbf{r}} V(\mathbf{r}).$$

Thus, the first Born approximation for the scattering amplitude in a quantization box is

$$f^{(1)}(\theta, \phi) = -\frac{2m}{\hbar^2} \frac{L^{3/2}}{4\pi} \frac{1}{L^3} \int_V d^3r e^{-i\mathbf{q} \cdot \mathbf{r}} V(\mathbf{r}).$$

Up to the overall factors associated with box normalization, the Born amplitude is the Fourier transform of the potential evaluated at the momentum transfer $\mathbf{q}$.

It is useful to also move into the continuum to get

$$f^{(1)}(\theta, \phi) = -\frac{m}{2\pi\hbar^2} \frac{1}{(2\pi)^3} \int d^3r e^{-i\mathbf{q} \cdot \mathbf{r}} V(\mathbf{r}).$$

The first Born approximation for spherically symmetric potentials

Now we move to cases where calculations are easier and which are usually extremely relevant. This is the special case of spherical symmetry of the potential (no dependence on θ and ϕ). As the scattering is elastic

$$q = |\mathbf{k} - \mathbf{k}'| = 2k \sin \frac{\theta}{2}.$$

This allows us to perform the angular integration explicitly to obtain

$$f^{(1)}(\theta) = -\frac{1}{2} \frac{2m}{\hbar^2} \frac{1}{iq} \int_0^\infty r^2 V(r) (e^{iqr} - e^{-iqr}) dr = -\frac{2m}{\hbar^2} \frac{1}{q} \int_0^\infty r V(r) \sin(qr) dr.$$

Let us apply this theory to two models typically used in scattering experiments in nuclear physics.

Name	Formula	Relevance	Range
Yukawa potential	$V(r) = -g^2 \frac{e^{-\mu r}}{r}$	nuclear force (pion exchange) screened Coulomb in plasmas effective interactions in colloids	short
Gaussian potential	$V(r) = V_0 e^{-r^2/a^2}$	cold-atom scattering smooth optical traps molecular / nuclear cluster models	short
Finite square well	$V(r) = \begin{cases} V_0, & r < a, \\ 0, & r > a \end{cases}$	semiconductor quantum wells / dots simple nuclear shell model impurities in solids	short
Hard-sphere potential	$V(r) = \begin{cases} \infty, & r < a, \\ 0, & r > a \end{cases}$	neutron–nucleus scattering molecular collisions s-wave scattering in cold atoms	short
Delta-shell potential	$V(r) = \lambda \delta(r - a)$	solvable resonance model thin nuclear surface pedagogical toy potential	short
Coulomb potential	$V(r) = \pm \frac{q_1 q_2}{4\pi\epsilon_0 r}$	hydrogenic atoms Rutherford scattering charged impurities in semiconductors	infinite

Table 1: Overview of common model potentials used in scattering theory and their relevance to different fields of research. The Coulomb potential requires special treatment; the naive Born approximation fails due to its long-range nature.

Example 1: Square well potential - model for elastic proton scattering of calcium isotopes

A simple but important example is scattering by a finite square well, which is a toy model for the process of elastic proton scattering from several nuclei, each of which is an isotope of calcium (the standard nucleus Ca^{40} has 20 protons and 20 neutrons). The potential is defined as

$$V(r) = \begin{cases} V_0, & r \leq a, \\ 0, & r > a. \end{cases}$$

which is a (soft) sphere kind of model (for infinite V_0 one calls this the hard sphere model). The dimension a quantifies the size of the nucleus. Adding more neutrons (making heavier isotopes) makes a larger.

Let us now evaluate the integral

$$\begin{aligned}
 f^{(1)}(\theta) &= -\frac{2m}{\hbar^2} \frac{1}{q} \int_0^\infty r V(r) \sin(qr) dr = -\frac{2m}{\hbar^2} \frac{V_0}{q} \int_0^a r \sin(qr) dr \\
 &= -\frac{2m}{\hbar^2} \frac{V_0}{q} \left[-\frac{1}{q} \int_0^a r \frac{d}{dr} \cos(qr) dr \right] = -\frac{2m}{\hbar^2} \frac{V_0}{q^2} \left([r \cos(qr)]_0^a - \int_0^a \cos(qr) dr \right) \\
 &= -\frac{2m}{\hbar^2} \frac{V_0}{q^2} \left(a \cos(qa) - \frac{1}{q} [\sin(qr)]_0^a \right) = -\frac{2m}{\hbar^2} \frac{V_0}{q^2} \left(a \cos(qa) - \frac{\sin(qa)}{q} \right) \\
 &= -\frac{2m}{\hbar^2} V_0 \left(\frac{a \cos(qa)}{q^2} - \frac{\sin(qa)}{q^3} \right).
 \end{aligned}$$

Slightly rewriting to express everything in terms of the dimensionless variable qa :

$$f^{(1)}(\theta) = -\frac{2m}{\hbar^2} V_0 a^3 \frac{1}{(qa)^2} \left(\frac{\sin(qa)}{qa} - \cos(qa) \right).$$

This function has zeros at

$$qa = 2ka \sin \frac{\theta}{2} = 4.49, 7.73, 10.9, \dots$$

Knowledge of the initial momentum and of the locations of these zero as the angle of observation is varied can be used to determine the well radius a - the size of the nucleus and therefore the isotope in the target.

Example 2: Yukawa potential in the Born approximation - model for the strong nuclear force

Another important example is scattering by a Yukawa potential

$$V(r) = V_0 \frac{e^{-\mu r}}{\mu r}$$

where V_0 is independent of r , and $1/\mu$ sets the range of the interaction, analogous to the parameter a in the square-well model. Notice that V goes to zero very rapidly for $r \gg 1/\mu$.

The Yukawa potential has a few applications among which we list here:

- Nuclear physics - this potential describes the short-range attraction between nucleons in a nucleus and originates from the exchange of *pions*. A pion is a light meson made of a quark–antiquark pair, with three charge states and a mass around $140 \text{ MeV}/c^2$. Because pions are massive, exchanging them produces a force with finite range $1/\mu$. This explains why nuclear forces are strong at femtometer distances but rapidly disappear beyond them, making the Yukawa form a simple and effective model of nucleon binding.
- In many condensed-matter and cold-atom systems, particles interact by exchanging massive excitations such as phonons or excitons, leading to Yukawa-type effective potentials. The finite mass of the mediator creates exponential screening, yielding short-range interactions. Such behavior appears in polaron physics, exciton-mediated electron interactions, and tunable interactions in ultracold atomic gases.

For this potential we obtain

$$\begin{aligned} f^{(1)}(\theta) &= -\frac{2m}{\hbar^2} \frac{1}{q} \int_0^\infty r \left(V_0 \frac{e^{-\mu r}}{\mu r} \right) \sin(qr) dr \\ &= -\frac{2mV_0}{\mu\hbar^2} \frac{1}{q} \int_0^\infty e^{-\mu r} \sin(qr) dr \\ &= -\frac{2mV_0}{\mu\hbar^2} \frac{1}{q} \Im \int_0^\infty e^{(-\mu+iq)r} dr \\ &= -\frac{2mV_0}{\mu\hbar^2} \frac{1}{q} \Im \left(\frac{1}{-\mu+iq} \right) \\ &= -\frac{2mV_0}{\mu\hbar^2} \frac{1}{q} \left(-\frac{q}{\mu^2+q^2} \right) = -\frac{2mV_0}{\mu\hbar^2} \frac{1}{q^2+\mu^2} = -\left(\frac{2mV_0}{\mu\hbar^2} \right) \frac{1}{4k^2 \sin^2 \frac{\theta}{2} + \mu^2}. \end{aligned}$$

So, in the first Born approximation, the differential cross section for scattering by a Yukawa potential is given by

$$\left(\frac{d\sigma}{d\Omega} \right) = \left(\frac{2mV_0}{\mu\hbar^2} \right)^2 \frac{1}{[2k^2(1 - \cos \theta) + \mu^2]^2}.$$

References

- [1] J. J. Sakurai and Jim Napolitano, Modern quantum mechanics <https://icourse.club/uploads/files/65fb4c16648d2b93f82fe3271c3381c211f2f532.pdf>
- [2] ScienceClic: <https://www.youtube.com/watch?v=oA3Sbm2PIGI&t=307s>